

Project 2: Morphological Image Processing

Given a binary image, please design a morphological algorithm to find its boundary. An example is shown in Figure 1.

To Do: Please process the image (size: 1024×1024) “\Figure_P2.jpg”. Save your output picture as *Output_P2.jpg*. Tips: you need to binarize the original image first.

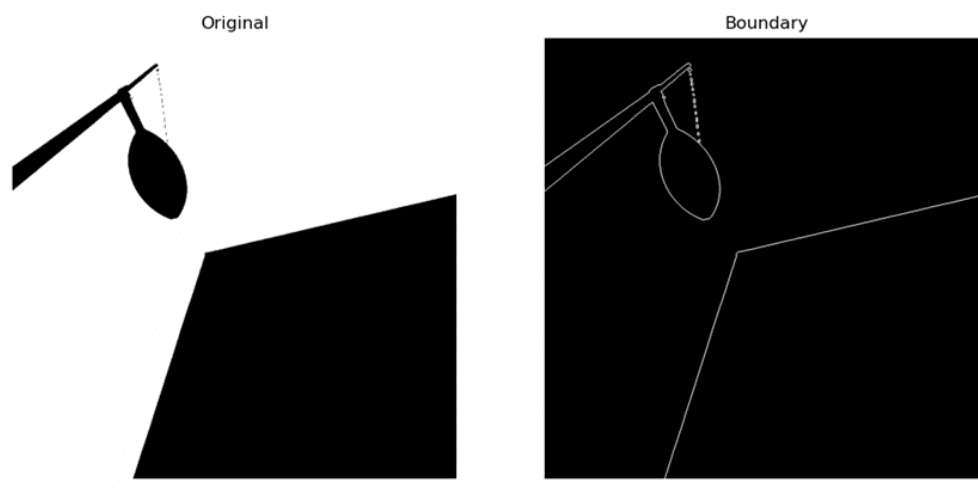


Figure 1. A binary image and its boundary (example, not same with Figure_P2).

Scoring Criteria:

- (1) The style, organization, and content of the report;
- (2) The correctness of the method used;
- (3) The performance of the code.

Note:

- (1) You can use any programming language you like.
- (2) You **cannot** call any third-party or built-in library for key processing steps. However, some basic functions such as image I/O, visualization or mathematical operations are allowed. Taking python as an example, you cannot call any function from cv2, but you can use I/O and visualization functions in cv2 and use numpy to do mathematics.
- (3) You are required to submit the source code and a report (a template is provided).
- (4) Please pack all files (code and report) into one compressed ZIP/RAR file named

‘proj2_student-id_short-name.zip/rar’, and send it to:
<https://send2me.cn/2E91DPbO/S7SRbJrxdddBGQ>

NOTE: the above submission link is DIFFERENT WITH the link for Project 1!!!

Please note that your score will be reduced or even set to zero if missed or wrong submission is found.

(5) Deadline: May 31, 2026.

(6) Do it by yourself! Plagiarism is strictly prohibited.